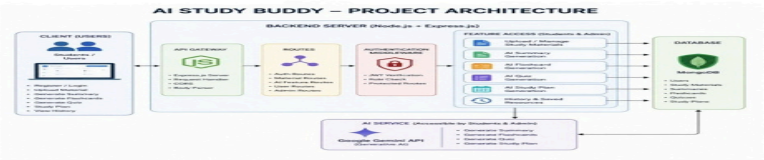
LEARNMATE – AI STUDYBUDDY

3. Project Design Phase

Designing the modular REST architecture, data model and application layers.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks

Epic 1: Project Architecture



The **AI StudyBuddy** backend follows a modular RESTful architecture designed to provide secure, scalable, and AI-powered learning services. The system processes client requests through Express.js, validates users using JWT authentication, interacts with the MongoDB database for data storage, and communicates with the Google Gemini AI API to generate intelligent study resources.

The architecture separates authentication, business logic, AI services, and database operations into independent modules, making the application easier to maintain, extend, and debug.

Components Definition

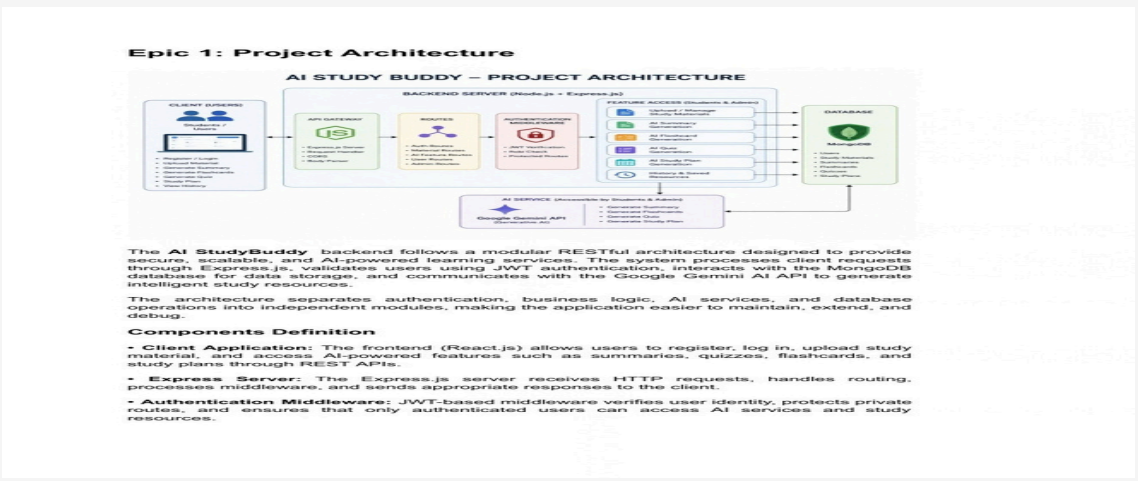
- **Client Application:** The frontend (React.js) allows users to register, log in, upload study material, and access AI-powered features such as summaries, quizzes, flashcards, and study plans through REST APIs.
- **Express Server:** The Express.js server receives HTTP requests, handles routing, processes middleware, and sends appropriate responses to the client.
- **Authentication Middleware:** JWT-based middleware verifies user identity, protects private routes, and ensures that only authenticated users can access AI services and study resources.

AI-powered learning assistance and smart study support platform

System Architecture

Key Points

- React.js client communicates with the Express.js server through REST APIs.
- JWT authentication middleware verifies identity and protects private routes.



Reference visual from the supplied LearnMate project document.

- Route handlers manage authentication, study materials, AI features and administrative operations.
- A dedicated AI service communicates with Google Gemini API.
- MongoDB with Mongoose stores users, study materials and AI-generated resources.

Data Design

Key Points

- User, Study Material, Summary, Flashcard, Quiz and Study Plan entities are represented as MongoDB collections.
- User → Study Material is one-to-many; Study Material → Summary is one-to-one.
- Study Material → Flashcards is one-to-many; Study Material → Quiz is one-to-one.
- User → Study Plan is one-to-many and all AI resources are associated with their user.

Architecture Pattern

Key Points

- The backend follows MVC architecture: Model for data, View as structured JSON API responses, and Controller for business logic.